

(100 points)

$$\mathbf{x} = [0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1]$$

Similarly, if the initial number in R0 is 0xCCCCCCCC (in binary, it is 1100110011001100110011001100; in decimal, it is equal to 3435973836), the array x after program completion will be:

$$\mathbf{x} = [0, 0, 1, 1, 0, 0, 1, 1, 0, 0, 1, 1, 0, 0, 1, 1, 0, 0, 1, 1, 0, 0, 1, 1]$$

```
length: .word 32
```

Note: You can demo this lab at the same time as Lab 6. Alternately, you can submit your code (as a .txt file, Canvas will not accept any other format) through the Assignment portal in Canvas. The Canvas submission window will close at noon, June 8, 2026. Please note that no grace period will be provided to finish your lab.

Make sure that your code does not have an infinite loop (identifiable by a persistent green bar at the top with the word “Running” toward the left after you click “Continue”; this means you have a logical error in loop exit). Codes with an infinite loop will receive no credit.